

SAMPATI ANVEKAR

Software Engineer | Full Stack (MERN) | AI/ML | Security & Testing | MCA Graduate

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PROFESSIONAL SUMMARY

MCA graduate from BMS College of Engineering, Bengaluru with hands-on experience in full-stack development (MERN), AI/ML, and application security and testing. Built a logistics optimization platform using Ant Colony Optimization, conducted OWASP Top 10 penetration testing, identifying and remediating 6 vulnerability classes, and proposed HybridNet — a custom DenseNet121-based architecture achieving 89.7% accuracy for AI-generated image detection in cyber forensics. Ranked #1 in BCA (92.25%).

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, C++

Web & Frameworks: React.js, Node.js, Express.js, HTML, CSS, REST APIs

Databases: MongoDB, MySQL, SQL, NoSQL

Tools & Platforms: Git, GitHub, Postman, Kali Linux, Arduino, AWS

AI/ML: Deep Learning, NumPy, Pandas, Linear & Logistic Regression, Grad-CAM

Security: OWASP Top 10, Penetration Testing, SQLi, XSS, Secure Code Review

Testing: Software Testing, Selenium, Manual & Automated Testing

OS & Methods: Windows, Linux, Agile, DSA, Source Code Revision Control

EXPERIENCE

Software Engineer Intern – AI/ML & Data Science | AIROBOSOFT PRODUCTS AND SERVICES LLP, Bengaluru | Mar–Apr 2025

- Gained hands-on experience in Python, NumPy, Pandas, and core ML algorithms, building and evaluating Linear & Logistic Regression models on a real-world sales prediction use case.
- Developed a Python-based personal assistant as a capstone project, integrating NLP concepts and modular software design principles.

PROJECTS

Fleet Route Optimization System | Python, MERN Stack, Ant Colony Optimization

- Built a full-stack logistics platform (MERN + Python) using ACO to compute and compare optimal delivery routes, with a real-time dashboard visualizing distance, ETA, and fleet performance metrics.
- Designed RESTful APIs connecting the Python optimization engine to the React frontend, demonstrating end-to-end cross-stack integration.

HybridNet – AI-Generated Image Detection | Python, Deep Learning, Cyber Forensics

- Proposed HybridNet, a custom DenseNet121-based architecture using 6-channel hybrid inputs (RGB + Residual + PRNU), achieving 89.7% accuracy — outperforming LightCNN, ResNet18, and Xception.
- Integrated Grad-CAM explainability for forensic interpretability; benchmarked against 3 state-of-the-art deep learning models.

Web Application Penetration Testing (OWASP Top 10) | Kali Linux, OWASP Top 10, Web Security

- Conducted end-to-end penetration testing on a live web application, identifying and exploiting 6 vulnerability classes including SQLi and XSS; authored a remediation report eliminating all critical findings.

Fitfinity – Health Platform | HTML, CSS, JavaScript, PHP, MySQL

- Engineered a multi-role health platform with doctor search, appointment booking, and BMI calculators; integrated Google Translate API for multilingual accessibility. Received Best Project Award.

Obstacle Avoidance Robot Car | Arduino, Embedded C, Bluetooth, Ultrasonic Sensors

- Built a Bluetooth-controlled robot with autonomous obstacle detection via ultrasonic sensors, real-time path adjustment, and multilingual voice command support.

EDUCATION

MCA – BMS College of Engineering, Bengaluru | 77.61% | 2023–2025

BCA (1st Rank) – KLE's RLSI College of BCA, Belagavi | 92.25% | 2020–2023

12th PUC – St. Joseph's PU College, Karwar | 89.88% | 2020

10th SSLC – The Central School, Dubai, UAE | 69% | 2018

ACHIEVEMENTS & CERTIFICATIONS

- Best Project Award for 'Fitfinity'
- Secured 1st Rank in UG (92.25%)
- Secured 'IT Manager' title at BMSIT inter-college event (2025)
- AWS Academy Graduate – Cloud Foundations certified